Ayoub TOUJANI

Computer Science Engineer: Data & Business Intelligence

Eligible for the Young Professionals Agreement (Switzerland-Tunisia)

25 years old | Tunisian nationality | Tunis, Tunisia
+216 99 356 653 | toujaniayoub808@gmail.com | Willing to relocate
linkedin.com/in/ayoubtoujani | github.com/toujaniayoub | Portfolio



PROFESSIONAL SUMMARY

Data and Business Intelligence Engineer with experience in data management, analytics, automation, and AI solutions within the banking and telecommunications sectors. Experienced in designing end-to-end data platforms, automating business processes, and developing dashboards that support operational and strategic decision-making. Specialized in **Data Engineering, Business Intelligence, Predictive Analytics, and Artificial Intelligence**. Available immediately, eligible for the Switzerland-Tunisia Young Professionals Agreement, fully mobile.

PROFESSIONAL EXPERIENCE

SIBTEL - Interbank Clearing Company of Tunisia Dec. 2025 – June 2026
Automation & Data Engineer *Tunis, Tunisia*

- Designed and deployed, the first decision-support platform at SIBTEL, the sole interbank clearing operator in Tunisia (**+50 million transactions/year**, 28 member institutions).
- Built a single geographic reference for **2,200+ bank branches** through **3 n8n workflows** merging 3 different sources (internal Oracle system, Central Bank API, national postal reference data), replacing a fully manual process and giving SIBTEL its first-ever geographic map of the network.
- Automated the daily collection, cleaning, and enrichment of data (duplicate/conflict resolution, geocoding), with zero manual work needed since going live.
- Orchestrated a daily **ELT** pipeline with **Apache Airflow** feeding a **star-schema** data warehouse, protected by **40 automated quality tests** that block any bad data before it reaches users.
- Built **6 self-service Metabase dashboards** for all bank members : *National Daily Overview*, *Bank Comparison*, *Agency Network* (geographic scoring of 2,200+ branches), *Payment Analytics*, *Bank & Agency 360 °*, and *District & Regional Performance* cutting analysis time from several days to under 15 minutes.
- Deployed a full **monitoring** stack (Prometheus, Grafana, 3 exporters, 8 alert rules) covering the pipeline, warehouse, database, and server, to keep the whole platform reliable end to end.
- Turned business needs from the Data Analytics, BI & Business Development unit into technical specifications and architecture choices, validated during weekly reviews with management.

Tunisie Telecom June 2025 – Sept. 2025
Business Intelligence Engineer *Tunis, Tunisia*

- Designed and deployed, a business intelligence solution for the Regional Operation Center, turning a fully reactive incident process spread across **52 monthly CSV files** (4 years of history) into one centralized, proactive view.
- Modeled (**Kimball** method, star schema: 1 fact table, 5 dimensions) and built a **PostgreSQL data warehouse**, fed by an **ETL pipeline in Talend Open Studio** that included SQL-based cleanup of the service and problem-type fields.
- Built **3 Power BI dashboards**: *Incident Management* (executive view and 4-year trends), *Problem / Root Cause Analysis* (spotting recurring causes), and *Site & Service Performance* (resolution time by site and priority) - the company's first-ever consolidated view of incident performance.
- Trained a **machine learning** model to predict how long an incident will take to resolve, based on its type, service, priority, and site, and exposed it through a **Flask API** to help plan resources ahead of time.
- Built an **Angular** web app bringing the Power BI dashboards and the prediction model together in one portal.

KEY PROJECTS

EDvance: Admission and Employability BI & AI Solution Jan. – May 2025 GitHub
Designed and implemented a complete business intelligence solution using Agile methodology, including a PostgreSQL star-schema data warehouse, Talend ETL pipelines integrating SQL and Excel data, predictive Python models for enrollment forecasting, and Power BI dashboards integrated into an Angular/Flask web application.

Airline Delay and Performance Analysis Oct. – Dec. 2024 GitHub
Developed a complete decision-support solution including a SQL Server data warehouse designed using a star schema, SSIS ETL pipelines for data extraction and transformation, and Power BI dashboards providing key performance indicators such as delays, cancellations, punctuality rates, and airline rankings using DAX measures.

SmartPredict : AI Smartphone Recommendation System Sep. – Dec. 2024 GitHub
Developed a machine-learning solution to help users select smartphones based on their needs and budget. Built predictive models to estimate smartphone prices, implemented a recommendation system, and created interactive Power BI dashboards for data visualization.

TECHNICAL SKILLS

Data Engineering	Apache Airflow (DAGs, scheduling), dbt (tests, incremental models), n8n, Talend Open Studio, SSIS
Data Warehousing	PostgreSQL, TimescaleDB, MySQL, Oracle, SQL Server, Star schema, Kimball
BI & Visualization	Power BI (DAX, Power Query), Metabase, Grafana + Prometheus, KPI reporting, data quality
Analytics & AI	Python (Pandas, scikit-learn), Machine Learning, Flask API, Angular, supervised/unsupervised learning
DevOps & Tools	Docker Compose, Git, Microsoft Power Apps, SAP Fiori, Tesseract OCR, Advanced Excel

EDUCATION

National Engineering Degree in Computer Science 2023 – 2026
Private Higher School of Engineering and Technology (ESPRIT) - Tunisia

Bachelor's Degree in Computer Science 2020 – 2023
ISAMM - University of Manouba, Tunisia

LANGUAGES

French: Fluent (C1) | **English:** Fluent (C1) | **Arabic:** Native